



Certificate

We hereby confirm that

Samir Schabel

has completed the intensive program

Data Science & AI

with 960 hours of programming practice (corresponds to 720 lessons) successfully.

The course has been delivered remotely.

Participation from 22.09.2025 until 26.01.2026

Franz Wilhelmy
Program Manager

Petra Kischkewitz
Managing Director

Anastasia Mikheeva
Lead Coach

Curriculum Data Science & AI

In 720 lessons the students have learned, discussed and practiced the following content in several tasks and projects:

Programming & Database Python

- pandas
- scikit-learn
- TensorFlow/Keras
- Visualization
- Matplotlib
- seaborn
- Plotly
- Altair
- SQL
- Unix
- Git
- GitHub

Exploratory Data Analysis EDA

- & Statistics
- Data Analysis
- Distributional Analysis
- General Statistics
- Data Visualization
- Distribution Plots
- Relationship Plots
- Geo Visualizations

Data Science Portfolio Projects

- EDA Project
- Predictive Modeling Project
- Final Project / Capstone (4 weeks)

Machine Learning Algorithms

- Supervised Learning
- PCA (Dimensionality Reduction) under Unsupervised learning
- Linear Regression
- Logistic Regression
- KNN
- Decision Trees
- Random Forest
- Ensemble Methods
- Unsupervised Learning
- Clustering
- Deep Learning
- Natural Language Processing
- Time Series

Evaluation & Performance Metrics

- Confusion Matrix
- Regression and Classification Metrics
- Error Analysis
- Model Tuning
- Bias-Variance Tradeoff
- Regularization
- Cross-Validation
- Optimization
- Gradient Descent
- Cost Functions

Machine Learning Engineering

- sklearn Pipeline
- MLflow

Prompt Engineering RAG

- AI Agents
- Transformers and LLMs

Collaborative Working & Social Learning

- Pair Coding
- Agile Workflow
- Daily Stand-Ups
- Retrospective
- Kanban Project Board
- Daily Class Review
- Team Work & Self-Organization Skills
- Career Coaching & Mentoring

Communication with Stakeholders

- Business Presentation
- Non-Technical Stakeholders
- Data Science Lifecycle Presentation
- Technical Stakeholders
- Stakeholder Review
- Compact & Time-Critical Project Alignment